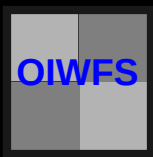


Near IR On-Instrument Wavefront Sensor (OIWFS)—Contents

Hubert Yamada

oiwfs_h ty_ TOC / 05-00 / 2001-07-02

This report contains a collection of tables of contents for the documentation for the Gemini Near Infra-Red On-Instrument Wavefront Sensor (OIWFS).



**niri_h ty_009.fm Near IR Imager (NIRI) and Wavefront Sensor (OIWFS)—Definitions, Abbreviations
and References**

1	Introduction	1
1.1	Purpose of the document	1
1.2	Scope	1
1.3	Abbreviations and Acronyms	1
1.3.1	Stylistic Conventions	2
1.4	Definitions	2
1.5	References	2

1	Introduction	i
1.1	Scope	i
2	Precautions	ii
2.1	Temperature	ii
2.2	Motor Heating	ii
2.3	Wear	ii
3	Procedure for Testing Mechanisms	ii
3.1	Preparation	ii
4	Filter Wheel	iii
4.1	Focus Stage	iii
4.2	Gimbal	iv

1	Introduction	1
2	Filter Wheel	1
3	Focus Stage.....	2
4	Gimbal Mirror	3

1	Introduction	1
1.1	Purpose	1
1.2	Definitions, Acronyms and Abbreviations	1
1.3	Stylistic Conventions.....	2
1.4	References.....	2
2	Release Notes	2
2.1	The 1.00 release (2000-06-27)	2
2.2	The 1.00 alpha 6 release (1999-11-11)	2
2.3	The 1.00 alpha 5 release (1999-10-23)	3
2.4	Release V003, Releases 1.00 alpha 1 - 1.00 alpha 4.....	3
2.5	Release V002.....	3
2.6	Release V001.....	3
3	Hardware configuration	3
3.1	VME Boards	3
3.2	Omega Temperature controllers	4
3.3	MV167 Real time clock	4
4	Prerequisites	4
4.1	VxWorks routing bug	5
4.2	Bug fix for e2sr (pre-GEM5).....	5
5	Building, Loading, and Using the NIRI Low-Level Engineering Software.....	6
5.1	Obtaining the Software	6
5.2	Unpacking the Software	7
5.3	Configuring a standard EPICS system	7
5.4	Configuration.....	7
5.4.1	Initial Setup	8
5.4.2	Environment subsystem	8
5.5	Building the software	8
5.6	Making dependency list.....	9
5.7	Loading the NIRI CC and OIWFS control software or simulator	9
5.8	Running the diagnostic software.....	9
5.9	Cleaning and rebuilding	9
6	Building, Loading, and Using the CICS/NIRI/OIWFS software	10
6.1	Obtaining the Software	10
6.2	Unpacking the Software	10
6.3	Configuring a standard EPICS system	10
6.4	Configuration.....	10
6.5	Building the system.....	13
6.6	Making dependency list.....	13
6.7	Loading the NIRI/CICS OIWFS/CICS control software or simulator	13
6.8	Running the user interface.....	13
6.9	Cleaning and rebuilding	14
7	Multiple copies of the software	14
8	Limitations of the simulation	14
9	Unfinished tasks	15
9.1	WCS	15
9.2	Networking and communications	15
9.3	wfsBeam	15

10	Known bugs and limitations	15
10.1	Problems with the simulation	16
10.2	Problems with edd / dm	16
10.3	Stability problems	17
10.4	Compiler Problems, Code, and Coding Standards	17
10.5	Mechanical problems	17
10.6	Datuming and diagnostic commands	18
10.7	Calibration tools	18
10.8	Other Problems	19
11	Possible Enhancements	19
12	Design Decisions (“It’s not a bug, it’s a feature!”)	20

1	Introduction	1
1.1	Purpose of the document	1
1.2	Scope	1
1.3	Definitions, Acronyms and Abbreviations	1
1.4	Stylistic Conventions	2
1.5	References	2
2	Implementation	2

1	Introduction	1
1.1	Purpose of the document	1
1.2	Scope	2
1.3	Definitions, Acronyms and Abbreviations	2
1.4	Stylistic Conventions	2
1.5	References	2
2	Implementation	2
2.1	Overview	2
2.2	Static Storage	4
2.2.1	Static Look-Up Tables	4
3	Stepper Motor / Hall-Effect Sensor Control	4
3.0.1	Notes on Figure 3	4
3.1	Hall-Effect Sensors and the Home Position	5
3.1.1	A discrete positioning mechanism (4-position filter wheel)	5
3.1.2	Continuous-positioning mechanisms (OIWFS focusing stage)	10
3.1.3	Other continuous-positioning mechanisms (science focus and gimbals)	12
3.1.4	Two position mechanisms (the pupil viewer)	14
3.2	Peak Finding Algorithm	16
3.3	Stepper-motor errors	16
3.4	Backup sensors	16
3.5	Hard Limits and Limit Switches	16
3.5.1	The focusing stages	16
3.5.2	The Pupil Viewer	16
3.5.3	The OIWFS Gimbal Mirror	17
3.6	Simultaneous motor motion	17
3.7	Stepper Motor Control Module Failure	17
3.8	Diagnose	17
3.9	Redatum	18
4	Environment Control	18
4.1	Cooling Motor	18
4.2	Temperature regulation	18
4.3	Heating Resistors	18
4.4	Temperature Sensors	18
5	Special Calibration Programs	19
6	The Gemini Interlock System	19
7	Special Epics Records	19
7.1	Modified records	19
7.1.1	Motor record (OMS VME8, VME44 device/driver)	19
7.2	Custom Records, Device Support, and Subroutine Records	19

1	Introduction	1
1.1	Purpose of the document	1
2	Overview	1
2.1	The ENG directory	2
2.1.1	The CC Database	2
2.1.2	The WFS Database	2
2.1.3	The GBL Database	3
2.1.4	The ENV Database	3
3	The Mech Symbol and Diagram	3
3.0.1	The Mech Symbol	3
3.1	External interfaces (the NIRI/CICS interface)	4
4	The HallStep Record and Symbol	6
4.1	Record processing	9
4.1.1	When an operation code is processed:	9
4.1.2	When the completion task is given a new operation	9

niri_h ty_004.fm	Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Calibration and Maintenance Tools
-------------------------	---

1	Introduction	1
1.1	Purpose	1
1.2	Definitions, Acronyms and Abbreviations	1
1.3	Stylistic Conventions.....	1
1.4	References.....	1
2	Running the tools	2
2.1	The Engineering User Interface	2
3	Scan Tool.....	2
3.1	Starting Scan Tool.....	2
3.2	Using Scan Tool	2
4	Cycle Tool.....	3
4.1	Starting Cycle Tool	3
4.2	Using Cycle Tool.....	3
5	Position Tool.....	3
5.1	Starting Position Tool	3
5.2	Using Position Tool.....	3
6	Calibration Procedure.....	3
6.1	The Science Focus Stage.....	3
7	Troubleshooting	4
8	Warnings.....	4

niri_h ty_008.fm	“A&G to NIRI/GNIRS On Instrument Wavefront Sensors”
-------------------------	--

Chapter 1	Introduction	3
1–1.0	Description	3
1–1.1	Purpose	3
1–1.2	Scope.....	3
1–2.0	Related Documents and Drawings	4
1–2.1	References.....	4
1–2.2	Definitions, Acronyms and Abbreviations	4
1–2.3	Stylistic Conventions.....	4
Chapter 2	Details of the A&G to NIRI/GNIRS OIWFS Interface	5
2–1.0	Introduction	5
2–2.0	Physical System Interfaces.....	5
2–2.1	Mechanical Interface	5
2–2.2	Optical Interface	5
2–2.3	Electronic Interface	5
2–2.3.1	NIRI	5
2–2.3.2	GNIRS / Standalone OIWFS	5
2–2.4	Mass/Balance	6
2–2.5	Thermal Interface	6

2-3.0	Software/Control Function Interface	6
2-3.1	Overview	6
2-3.1.1	Communications infrastructure	6
2-3.1.2	Software architecture	6
2-3.2	Behavior	6
2-3.3	Implementation	6
2-3.3.1	Command (CAD/CAR) records	7
2-3.3.2	Status Information Records	9
2-3.3.3	Position names	16
2-3.4	Detailed Command Description	17
2-3.4.1	Sequence Commands	17
2-3.4.2	General Engineering Commands	18
2-3.4.3	OIWFS Standard Mechanisms	18
2-3.5	Detailed Status information	21
2-3.5.1	General records	21
2-3.5.2	Component Status Records	21
2-3.5.3	Component Health Records	23
2-3.6	Alarm conditions	24
2-3.7	Debugging and Maintenance	25
2-3.7.1	Debugging Modes	25
2-3.7.2	Calibration	25
2-3.8	Simulation	25
2-4.0	Safety Issues	25

niri_h ty_014.fm	Near IR Imager (NIRI) and OIWFS — OIWFS Component Position Names
-------------------------	---

1	Introduction	1
1.1	Purpose	1
1.2	Definitions, Acronyms and Abbreviations	1
1.3	Stylistic Conventions.	1
1.4	References.	2
2	Tables	2
2.1	Filter Wheel.	2

niri_h ty_005.fm	“NIRI/GNIRS Science Instrument to On Instrument Wavefront Sensor”
-------------------------	--

Chapter 1	Introduction	3
1–1.0	Description	3
1–1.1	Purpose	3
1–1.2	Scope	3
1–1.3	Differences between NIRI and GNIRS.	3
1–2.0	Related Documents and Drawings	3
1–2.1	References.	3
1–2.2	Definitions, Acronyms and Abbreviations	4
1–2.3	Stylistic Conventions.	4
Chapter 2	Details of the NIRI/GNIRS to OIWFS Interface	5
2–1.0	Introduction	5
2–2.0	Physical System Interfaces	5
2–2.1	Mechanical Interface	5
2–2.2	Optical Interface	5
2–2.3	Electronic Interface	5
2–2.4	Mass/Balance	5
2–2.5	Thermal Interface	5
2–3.0	Software/Control Function Interface (wavefront sensor).	5
2–3.1	Overview	5
2–3.1.1	Communications infrastructure	5
2–3.2	Implementation	5
2–3.2.1	Command (CAD/CAR) records.	5
2–3.2.2	Status Information Records	8
2–3.2.3	Detailed SIR record description.	9
2–4.0	Software/Control Function Interface (Components Controller).	10
2–4.1	Overview	10
2–4.1.1	Communications Infrastructure	10
2–4.2	Implementation	10
2–4.2.1	Monitoring an external record.	10
2–4.2.2	Status Information Records	11
2–5.0	Safety Issues	11
2–6.0	Activity Completion	11

1	Introduction	1
1.1	Purpose	1
1.2	Definitions, Acronyms and Abbreviations	2
1.3	Stylistic Conventions.....	2
1.4	References.....	2
2	Warnings and General Information	2
2.1	Warnings (Very Important!)	2
2.2	A Few Notes About Edd/Dm Interfaces	3
3	General User Commands	3
3.1	Starting the User Interface.....	3
3.2	The Introduction Screen	3
3.3	The Components Controller Main Screen.....	3
3.4	The On-Instrument Wavefront Sensor Main Screen	3
4	Restarting After the IOC Has Been Rebooted	3
4.1	Wait for initialization to complete.....	3
4.2	Restart screens.....	3
4.3	Look for RED	3
4.3.1	Set the Temperature Interlock	3
4.3.2	Start the Closed Cycle Coolers (if needed)	3
4.4	Look for YELLOW.....	3
4.4.1	Datum Mechanisms	3
5	Advanced User commands	3
5.1	Starting the Low-Level Engineering Interface	4
5.2	The Main Engineering Screen.....	4
5.3	The Low Level Mechanism Control Screen	4
5.4	The Temperature Control Screen	4
6	Debugging tools	4
6.1	The Digital Input/Output Monitor (Dio)	4
6.2	The Analog Input Monitor (Atod).....	4

